

Compression workflow · warmed loop

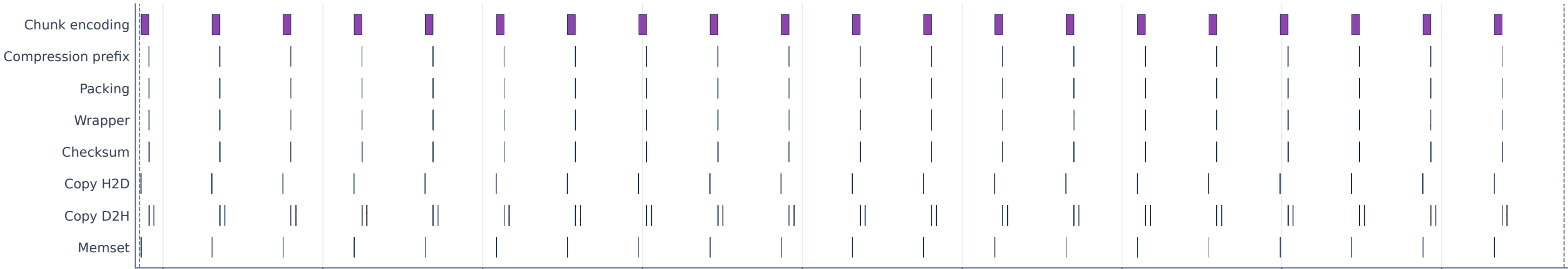
NVIDIA GeForce RTX 4090 · 64 MiB float32 payload · 20 validated measured iterations · source 7d083ff804e0

Instrumented diagnostics · 2 warmups · exact CUDA intervals and sampled process resources share elapsed time

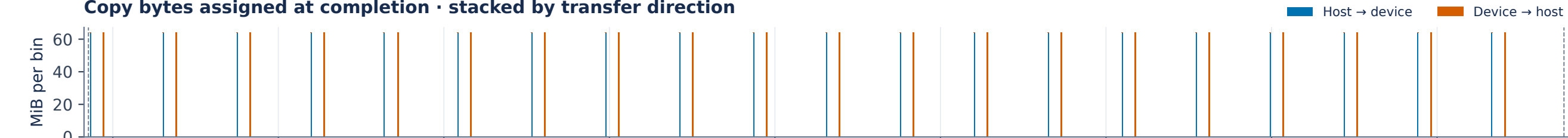
Host annotations · containers and nested download/check phases



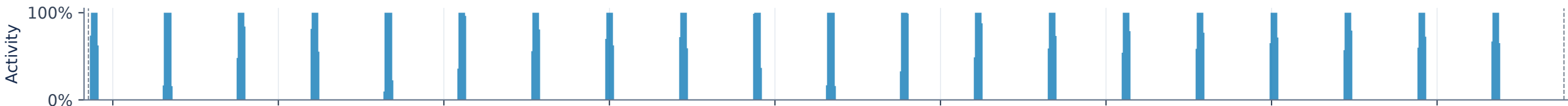
Exact recorded CUDA intervals · compression stages



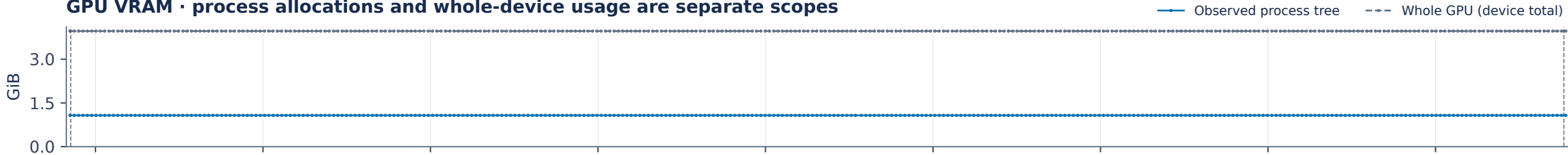
Copy bytes assigned at completion · stacked by transfer direction



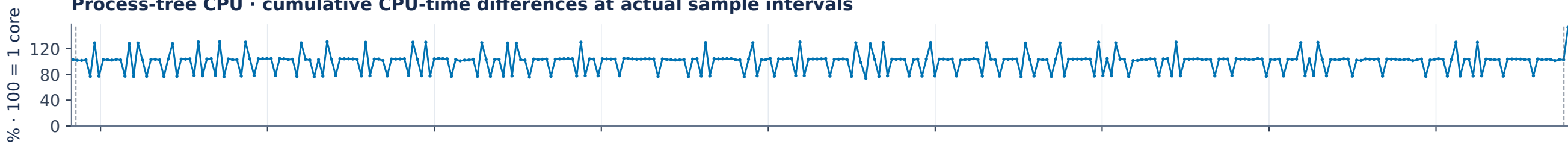
Recorded kernel activity fraction · interval union / bin duration; not occupancy



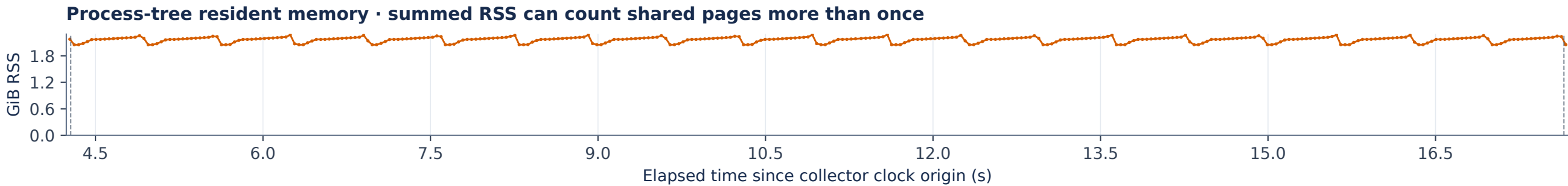
GPU VRAM · process allocations and whole-device usage are separate scopes



Process-tree CPU · cumulative CPU-time differences at actual sample intervals



Process-tree resident memory · summed RSS can count shared pages more than once



Dashed lines bound the measured loop. CUDA bins: 10 ms; view-boundary bins use their actual clipped width. Copies show completed bytes, not bandwidth.

Collector cadence: median 38.654 ms (37.735–45.628 ms). Metric points use query midpoints; CPU uses averaging-interval midpoints.

Host/CUDA alignment: $\pm 44.314 \mu\text{s}$ (half the origin-mark bracket). Host ranges can nest; they do not establish exclusive GPU ownership or costs.

Download, status validation, host byte construction and output validation are distinct host phases. Coarse resources cannot resolve individual short phases; nulls remain gaps.

Nsight warns potentially missing NVTX events; all expected host phase ranges were validated. Other warnings remain in the report and manifest.